High Power Boost 250W LED Driver / Charger Module 6A Input 8.5V-48V Output 10-50V Aluminium Substrate- A1401220



Link: https://robokits.co.in/power-supply/dc-dc-power-supply/high-power-boost-250w-led-driver-charger-module-6a-input-8.5v-48v-output-10-50v-aluminium-substrate--a1401220?srsltid=AfmBOorhqQJb9bXIF7hJj6TkLRwO2MbJ1pOH4Mkgt0Y74UcQY3MteT4-&

Features

- Module properties: non-isolated boost module
- Input reverse connection protection: Yes (less than 5A)

Specifications

- Input voltage: DC 10V~40V
- Input current: 6A (MAX)
- Static operating current: 10mA (when 12V step up to 20V, the higher the output voltage, the static current will increase)
- Output voltage: DC 10~50V, can be continuously adjustable (default output is 19V)
- Output current: 6A MAX, when exceeds 4A, please strengthen the heat dissipation (related to input and output pressure difference, the larger the pressure difference is, the smaller the output current is)
- Constant current range: 0.2~6A
- Output power: = Input voltage * 6A, for example: input 12V * 6A = 72W, input 24V * 6A = 144W, input 36V*5A=180W, input 48V*5A=240W
- Working temperature: -40~+85℃ (enhance heat dissipation if the ambient temperature is too high)
- Working Frequency: 150KHz
- Conversion efficiency: 92%~96% (efficiency is related to input, output voltage, current and differential pressure
- The output voltage / current regulation method: Adjust RV1 potentiometer, according to your battery or LED, set the output voltage to the voltage you need. For example, for 10 series LED voltage, regulate the voltage to 37V.
- Turn the RV2 potentiometer clockwise about 30 turns, set the output current to the minimum, connect the LED, and adjust the RV2 potentiometer to the current you need. For battery charging, after the battery is fully discharged, connect it to the output and adjust the RV2 to the current you need. (For charging, please make

sure the battery is fully discharged, then adjust the voltage value will be accurate, for the more battery left capacity, the smaller the charging current is.)

- Note: Do not use a short circuit method to adjust the current, the circuit structure of the boost module cannot be adjusted in a short circuit way.

Instructions for use

- VIN+ is the input voltage positive side, VIN- is the input voltage negative side.
- OUT+ is the output voltage positive side, OUT- is the output voltage negative side.
- Adjustable constant voltage range: DC 10~50V (clockwise regulation to decrease, counterclockwise regulation to increase).
- Adjustable constant current range: 0.2A~6A (clockwise regulation to decrease, counterclockwise regulation to increase).
- When the input current exceeds 2A, it is necessary to enhance heat dissipation (the aluminium substrate is attached to the flat metal part).
- The input voltage cannot be lower than 10V or more than 40V.
- The maximum input current cannot exceed 6A, the power cannot exceed 250W.
- Must ensure that the input voltage is lower than the output voltage at least 2V.
- Before connecting the load, please adjust the required voltage and current at no-load (no-load voltage should be increased by 3~5V at constant current mode).
- When using a switching power supply or other secondary power supply, please ensure that the power supply power should be double more than the power of the module.

Precautions

- The input and output cannot be reversed connection.
- The input and output voltage, the current must not exceed the maximum value during work to avoid damage to the boost converter module.
- When working with full load for a long time, pay attention to ventilation and heat dissipation to extend the life of the boost converter module.

Application

- Solar panel voltage regulator.
- Charge for the storage battery, lithium battery etc.
- Drive high-power LED lights.